



From Smartphone 3D to Federated Knowledge Graphs: Best Documentation Practices for Making Citizen-Generated 3D Heritage FAIR

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Abstract

Low-cost smartphone-based photogrammetry enables Citizen Scientists to generate 3D models of Cultural Heritage objects. However, these models are typically distributed as binary geometry files (e.g., OBJ, GLB) with minimal structured metadata, making them difficult to integrate into Linked Open Data (LOD) environments and Research Infrastructures.

The objective is to formulate a reusable documentation pattern aligning community-based 3D acquisition workflows with FAIR Digital Object (FDO) and FDOx principles and Semantic Web infrastructures (Fig. 1).

LOD has been established in Cultural Heritage to interlink dispersed datasets and publish semantically structured information in knowledge hubs such as Wikidata (Schmidt, Thiery & Trognitz, 2022). Recent initiatives emphasise community-driven LOD practices to integrate research data into distributed graph-based infrastructures (Thiery, Schenk & Thiery, 2025). In parallel, the FDO paradigm proposes packaging digital artefacts as persistent, typed, machine-actionable units enriched with structured metadata (De Smedt et al., 2020; Schwarzmänn, 2020). Recent work has explored integrating FDOs into federated knowledge graph environments (Fischer et al., 2025; Thiery, 2026a; Thiery, Noback et al., 2026; Thiery, Kasten et al., 2026) and provided reference implementations (Thiery, 2026b, 2026c). However, applying these concepts to citizen-generated 3D heritage and aligning them with community data hubs remains insufficiently addressed.

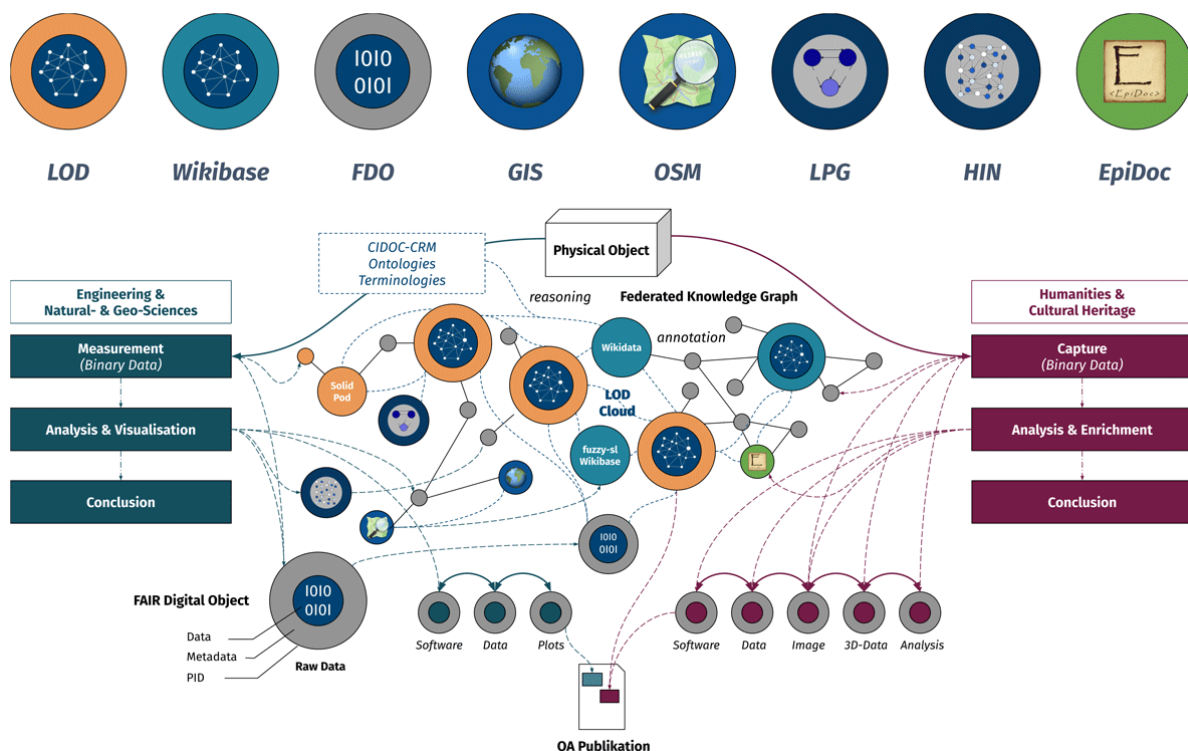


Fig. 1 Scheme of the Federated Knowledge Graph Ecosystem including FDOs, LOD and Wikibase instances

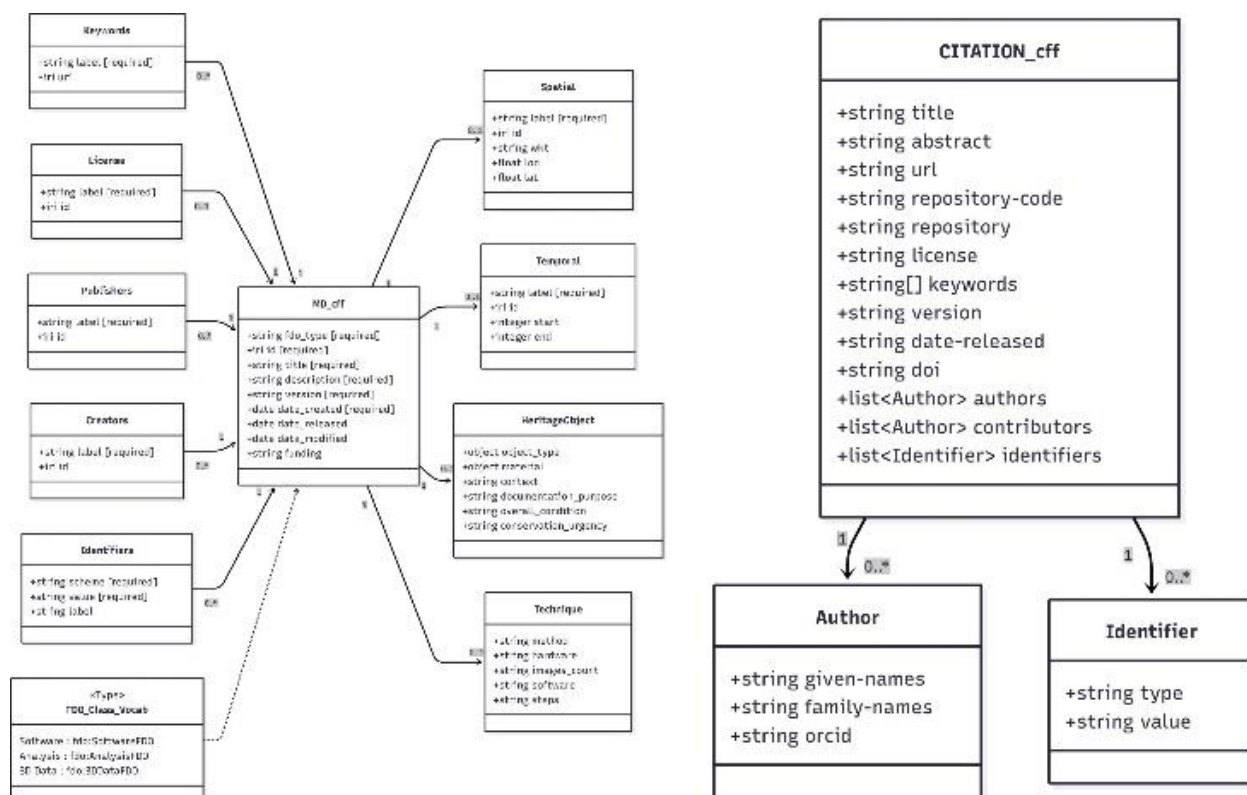


Fig. 2 The FDOx Metadata Scheme. Florian Thiery, CC BY 4.0.

The approach comprises a four-step documentation pattern:

- **FDO Encapsulation:** Each 3D model is encapsulated as an FDO with a persistent identifier, structured metadata, and explicit provenance. Binary geometry files remain unchanged, while metadata is exposed via standard schemes (e.g., DCAT, PROV-O).
- **Semantic Metadata and Paradata Modelling:** Acquisition parameters, spatial anchoring, and methodological details are modelled as paradata; ontology alignment (e.g. CIDOC CRM) supports interoperability. For the Ogham stone case study, this includes smartphone acquisition parameters, OSM-based spatial anchoring, and linking the inscribed monument to its documentation events (Fig. 2).
- **Linking to Community Hubs:** 3D models are linked to Wikidata items and OpenStreetMap entities via persistent identifiers for cross-referencing within the broader LOD Cloud.
- **Federated Knowledge Graph Integration:** RDF representations generated from the FDO layer enable participation in distributed infrastructures without centralising binary data.

The approach is illustrated using citizen-generated case studies (Fig. 3 & 4), including Irish Ogham stones and Holy Wells (Thiery, Distel & Thiery, 2025; Thiery, Schenk, Thiery & Thiery, 2025).

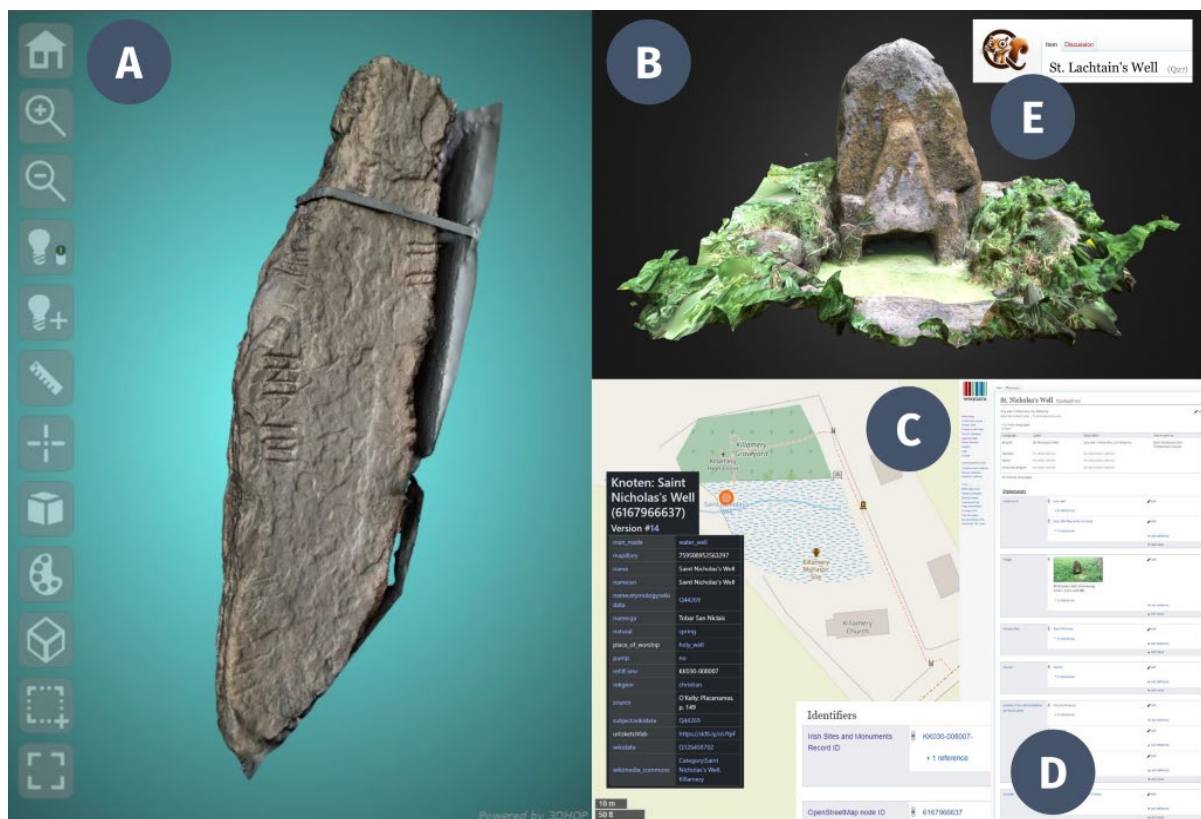


Fig. 3 (A) CIIC 81 Ogham Stone in 3D-HOP; (B) St Lachtain's Well in Sketchfab / OSM (C), in Wikidata (D) and within the SquirrelBase (E)

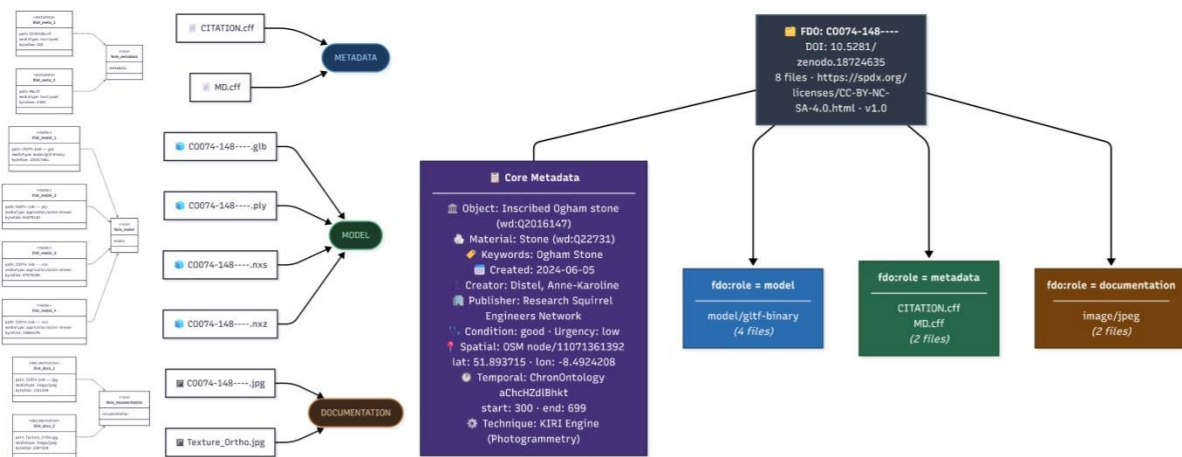


Fig. 4 CIIC 81 KIRI Engine SfM Scan as FDOx package.

The results demonstrate that citizen-generated 3D models become: (i) FAIR and citable research artefacts; (ii) semantically queryable entities within RDF environments; (iii) interoperable nodes connected to Community Hubs, and (iv) integrable components of federated knowledge graph infrastructures. The pattern enables structured metadata exposure, semantic linking, and sustainable referencing, while preserving native 3D formats. Key challenges concern scaling across heterogeneous acquisition setups and sustaining persistent identifiers and metadata quality as the contributor base grows.

The contribution combines best documentation practices, FAIR-compliant (meta)data modelling, ontology-aligned patterns, and community-driven interoperability for low-cost 3D heritage. Unlike generic FAIR and LOD practices, its novelty lies in coupling FDO encapsulation with community-hub linking, tailored explicitly to citizen-generated 3D heritage. By bridging low-cost 3D acquisition and infrastructure-level semantic integration, the paper proposes a scalable strategy for transforming binary 3D files into sustainable, machine-actionable components of Cultural Heritage ecosystems.

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